

Shadow-PPOV: Model-Free Passive Execution via Order-Level Shadowing and Identifier-Driven Cancellation

Vincent Maciejewski
M2 Technologies
mayeski@gmail.com

September 2026

Abstract

Automated execution algorithms fall into two broad families [9, 10]: schedule-based algorithms and liquidity-seeking algorithms. This paper concerns the schedule-based family; the liquidity-seeking family is out of scope. The schedule-based family commonly includes four algorithms: Time-Weighted Average Price (TWAP), Volume-Weighted Average Price (VWAP), Percentage of Volume (POV), and Implementation Shortfall (IS). TWAP follows a clock, whereas VWAP follows a forecast of the intraday volume *curve*, POV reacts to *realized* volume to hold a target participation rate, and IS front-loads execution to minimize slippage against the parent order arrival price. All operate on an *aggregate* (time, or total volume) rather than on the individual resting orders that actually constitute the book, and all are *model-based*: their execution decisions are derived from an explicit model, forecast, schedule, or control rule rather than from following individual resting orders. We introduce SHADOW-PPOV, a method for placing the passive limit orders within the framework of percentage-of-volume execution (a passive POV, or PPOV — here denoting passive execution whose *order-placement rate* is set from observed order flow rather than from traded volume; unlike conventional POV, that fraction is applied to individual order arrivals rather than to traded volume, and the participation rate is not controlled directly but emerges from the resulting fills). Placing a passive order to fill efficiently conventionally involves an order-book model and some prediction of where the order will fill. SHADOW-PPOV needs neither: it replaces that prediction with *tracking*, resting each passive order at the price of an order already in the book. It *shadows* the passive limit orders that other participants actually post: on observing a third-party add for a limit order, it may transmit its own limit order at the same price *on the same venue*, and it records only a single association between the observed order’s exchange identifier and its own shadow-order identifier. Cancellation is then *identifier-driven*: a shadow-order is withdrawn when the order it follows ends — at once when that order is cancelled, and after a brief grace window when it is hit by a trade. As this approach is model-free — it uses no order-book model and no fill prediction — it can serve as a benchmark for passive-order placement on central limit order book (CLOB) venues. We have evaluated SHADOW-PPOV over 137 replayed Chicago Mercantile Exchange (CME) futures sessions in a deterministic market-replay simulator, where it executes with slippage slightly worse than the mid-price reference. We position SHADOW-PPOV as a model-free benchmark for passive-order placement, against which a predictive placement model can be scored. The algorithm and the order-book simulator are implemented in the open-source KASPAR-HFT project.¹

Keywords: optimal execution, percentage-of-volume, adverse selection, limit order book, smart order routing, market microstructure, cancel mirroring.

¹<https://github.com/vincent212/kaspar-hft>

1 Introduction

A large order cannot be executed in a single trade without paying an impact cost proportional to how much liquidity it consumes at once. The standard response is to slice the parent order into child orders spread over time. The question every execution algorithm must answer is *how to slice*: how much to trade, when, and at what price.

TWAP commits to a clock: it slices evenly in time, blind to where liquidity actually is. But it must decide how much of each slice to work passively rather than cross the spread, and that tactical choice is shared across schedule-based algorithms rather than tied to any one of them [16, 17]. VWAP commits to a *forecast of the intraday volume curve* and weights its slices to match it; the forecast error is the tracking error. Percentage-of-volume (POV) has a participation ratio *reactive* to realized volume — it does not forecast a volume curve the way VWAP does. But POV cannot participate *passively* by reaction alone, because volume that has already printed can no longer be joined at a resting price; to stay passive it must lean slightly ahead of near-term volume (or cross the spread to catch up), and its completion time depends on volume it cannot yet see. So POV trades a *volume-curve* forecast for a short-horizon anticipation plus completion uncertainty, and it tracks an *aggregate* volume statistic rather than the individual resting orders that make up the book.

Implementation Shortfall takes a different benchmark: the arrival price, the market price at the moment the parent order is released. It measures execution cost as the gap between that decision price and the realized average fill price, including the opportunity cost of any unfilled quantity. Minimizing the gap trades market impact, which favors slow trading, against timing risk from price drift and volatility, which favors fast trading. The resulting schedule is front-loaded and tuned by a risk-aversion parameter: higher risk aversion trades faster to reduce exposure to adverse drift, at the cost of more impact. Like the volume-based algorithms, IS trades to a schedule computed in advance from a volume and impact forecast. Perold [11] introduces the IS formulation, and Almgren–Chriss [1] give the arrival-price optimal-execution formulation IS approximates.

This work concerns *passive* execution specifically, so the way POV places passive orders is central. A conventional POV algorithm is a hybrid: it rests passive limit orders when the market allows, but to hold its participation rate it also sends marketable orders, and it crosses the spread to catch up when it falls behind or when a slice deadline arrives [12, 14]. The passive-versus-aggressive choice is not special to POV: schedule-based implementations commonly face the additional tactical question of how much of each slice to work passively rather than cross, and this tactical layer is shared across schedules rather than tied to any one algorithm [16, 17]. A schedule that crossed on every slice would pay the spread needlessly, so even TWAP typically posts each slice passively and crosses only the unfilled residual near the slice deadline [18].

Wherever a slice is worked passively, the same placement problem arises: how far into the book to quote, trading fill probability against market impact and adverse selection [13, 22, 15]. A passive order can simply be posted at the touch, which needs no model; but posting blindly at the touch surrenders control of that trade-off and is prone to adverse selection — the order fills exactly when the level is thinning ahead of an adverse move. The conventional solutions instead build an order-book model and predict where a passive order will fill: optimal placement of limit versus market orders [12], optimal liquidation with limit orders under a fill-intensity model [13], and convex optimization of order placement across the book [14]. SHADOW-PPOV removes the need for an explicit order-book model and fill prediction from the placement decision: it resolves the placement problem by *following*, resting each passive order at the price of an order that just arrived at the book. This replaces order-book modeling and fill prediction with tracking of where liquidity actually rests. SHADOW-PPOV governs the placement of passive orders only; how much of a parent order to work

passively versus aggressively is a separate scheduling decision. Production implementations may mix passive and aggressive orders, and this paper makes no claim that aggressive orders are inferior. Its model-free character sets SHADOW-PPOV apart from standard execution algorithms, and lets it serve as a baseline against which predictive placement models can be measured (Section 10).

Beyond the question of how much to post and when, conventional execution stacks carry two further sources of overhead that this work targets directly:

- **Routing computation.** Smart order routing (SOR) across venues is normally a calculation: aggregate per-venue book depth, compute venue attractiveness scores, model per-venue latency, price maker/taker fees, and optimize a routing objective. The routing decision itself becomes a latency-bearing bottleneck.
- **State-dependent cancellation.** Order-management systems typically decide whether to pull a resting order by continuously evaluating market state — spread, queue position, fill probability, price competitiveness — which requires maintaining substantial per-order state and adds decision latency.

The premise of this paper is that all three — the forward placement of passive orders, the routing computation, and the state-dependent cancellation — are avoidable for a passive execution mandate, and that avoiding them yields a simple, computationally light method. The algorithm does not need to predict anything; it only has to follow the market. SHADOW-PPOV replaces the forward commitment with an observation: the passive orders other participants have *already committed to the book*. In the price-time-priority (FIFO) model used here, a limit order placed at the same price and venue as a resting participant order joins behind that order in the queue; a sweep must therefore consume the participant order and the depth ahead of the shadow before reaching the shadow. The method replaces the routing calculation with a pass-through of the observed order’s originating venue, and replaces state-dependent cancellation with an identifier rule: the shadow is withdrawn when the order it follows ends.

Contributions.

1. We propose SHADOW-PPOV as a simple, *model-free* benchmark for passive-order placement, built on *participant shadowing*: on observing a third-party add, SHADOW-PPOV places a passive order at that add’s price and venue, attaches it to the arriving order by exchange identifier, and ties its lifecycle to that order. Placement uses no order-book model and no fill prediction, giving predictive placement models a reference to beat (Sections 6, 10).
2. We define *identifier-driven cancel mirroring*: cancellation triggered by a delete/trade for the followed order’s identifier rather than by market-state evaluation, retaining a single identifier association per followed order — the exchange order identifier paired with the shadow’s identifier — so the cancellation decision is an $O(1)$ expected-time identifier lookup, independent of the number of tracked shadows (Sections 6.5–6.6).
3. We show that venue selection is *flow-inherited*: placing on the originating venue of each shadowed add makes the execution venue distribution match the observed add distribution without any venue score or routing optimizer — an inherited split rather than an optimized one (Section 6.7).
4. We describe the *reference implementation* in the KASPAR-HFT system, which layers state-based safety guards (a two-band place/cancel hysteresis plus completion and over-hedge guards) atop the core method, distinguishing the minimal core from the implemented superset (Section 7).

5. We report a measured evaluation over 137 replayed NQ/ES sessions, run through the KASPAR-HFT deterministic replay simulator, explicit about the split between the close-auction and open-auction measurements (Section 8).

2 Prior art: existing execution methods

Automated execution in current practice is organized around two broad families: schedule-based methods and liquidity-seeking methods [9, 5, 21]. We describe each operationally and note how SHADOW-PPOV differs; the academic results underpinning these methods are reviewed separately in Section 3, and the closest patented mechanisms in Section 4.

Scheduled slicing (TWAP, VWAP). The most widely deployed methods slice a parent order onto a schedule. TWAP spreads child orders evenly in time. VWAP shapes the schedule to a forecast of the intraday volume curve so that the realized average price tracks the volume-weighted benchmark. Both commit, before trading, to a trajectory keyed to a clock or a predicted volume profile, and both must post ahead of where volume will be in order to participate. SHADOW-PPOV commits to no such trajectory: it places only in reaction to orders already resting in the book, and does not forecast or pre-position against future volume. Kissell [5] covers both TWAP and VWAP in detail.

Percentage-of-volume (POV). A POV algorithm holds a target participation rate in market volume, scaling its trading up and down with *realized* volume rather than a fixed clock. It does not forecast a volume curve, but it does track an *aggregate* statistic (total traded volume) and, to participate passively, must lean slightly ahead of near-term volume rather than cross the spread. SHADOW-PPOV is POV-like in effect — its trading scales with the rate of observed adds — but its participation emerges from following *individual* resting orders at a bounded rate, and it does not pre-position against future volume. Kissell [5] covers POV in detail.

The POV control loop, and how Shadow-PPOV differs. A conventional POV algorithm runs a feedback loop against an aggregate statistic. Let ρ be the target participation rate, V_t the cumulative market volume, and x_t the quantity executed so far. At each step the algorithm compares x_t against the target ρV_t : when $x_t < \rho V_t$ it trades more, posting aggressively or crossing the spread to catch up; when $x_t > \rho V_t$ it slows or pauses.

SHADOW-PPOV replaces this control loop with per-order following (Section 6). Its actions are driven by the arrival of individual orders: on each observed add it may post a shadow at that order’s price and venue, bounded by that order’s size. The shadow rests just behind a live order at the same price, so under FIFO priority a sweep must consume the followed order and the depth ahead of the shadow before reaching it. Participation emerges from the rate of order arrivals, placement and venue selection are inherited from the followed order, and a forecast of forward volume is unnecessary. Table 1 contrasts the two mechanically.

Smart order routing (SOR). When liquidity is fragmented across venues, SOR decides where to send each child order by computing and comparing per-venue liquidity, fees and rebates, fill rates, and latency, and optimizing a routing objective. SHADOW-PPOV performs none of this: it sends each shadow to the originating venue of the add it follows, so the venue mix is inherited from observed order flow (Section 6.7) rather than solved for.

Table 1: Conventional POV and SHADOW-PPOV, by mechanism.

Aspect	Conventional POV	SHADOW-PPOV
Quantity tracked	aggregate market volume V_t	individual resting orders
Participation	target rate ρ , held by a feedback loop	emergent from following adds at a bounded rate
Passive placement	leans ahead of forecast volume	mirrors orders already resting
When behind	crosses the spread, paying impact	following continues until the target size is filled
Book / venue placement	separate SOR and queue logic	inherited from the followed order
Fill rate	depends on placement quality	requires a sweep that clears the followed order first (shadow rests just behind)

Liquidity-seeking algorithms. A liquidity-seeking algorithm aims to complete an order by capturing whatever liquidity is available at acceptable prices, prioritizing completion and low information leakage over any schedule or benchmark. As the second family of execution methods, it abandons a fixed schedule and sources liquidity opportunistically: posting and taking across lit and dark venues, and using hidden and iceberg orders and immediate-or-cancel probes to find undisplayed block size [5, 21]. It raises participation when liquidity appears at an acceptable price and withdraws otherwise, and may finish early when a block is found or wait when nothing suitable appears. SHADOW-PPOV operates on CLOB venues only, following the *displayed* resting orders on the book. It does not route to dark pools and makes no attempt to detect hidden or iceberg liquidity — it follows what the book shows, and commits to no trajectory of its own. SHADOW-PPOV and a liquidity seeker pursue different goals: a liquidity seeker minimizes footprint and hunts for size across lit and dark venues; SHADOW-PPOV places a passive order well on the lit book once a schedule has already decided to rest one. The two families are complementary — a liquidity seeker decides *when* enough liquidity is present to act, whereas SHADOW-PPOV decides *where* to rest a passive order.

3 Related literature

The methods above rest on, and are analyzed by, a substantial academic literature; we review the strands most relevant to SHADOW-PPOV’s design choices.

Optimal-execution theory. The modern foundation is Almgren–Chriss [1], which casts execution as minimizing a mean–variance trade-off between market impact (favoring slow trading) and timing/volatility risk (favoring fast trading), yielding an optimal schedule given models of permanent and temporary impact. A large body of work refines this framework, including limit-order formulations with adverse selection and fill uncertainty in the spirit of Avellaneda–Stoikov [2]. This theory answers *how fast* to trade given an impact model and a volume profile; it presupposes the profile. SHADOW-PPOV keeps the passivity these analyses target but does not compute a trajectory — it follows one the market reveals.

Queue position and adverse selection. Under price–time priority, both fill probability and adverse-selection cost depend on where a resting order sits in the first-in, first-out (FIFO) queue. Moallemi–Yuan [19] give a valuation model in which adverse-selection cost *increases* with queue position, with a dynamic component capturing the option value of locking in a position early. This literature quantifies precisely the effect SHADOW-PPOV is designed around: leading the book — posting early and deep in a thin queue — buys priority but courts selection when the level thins ahead of an adverse print. By posting alongside an order a participant has already chosen to rest, SHADOW-PPOV avoids deliberately taking priority ahead of that order. The followed order therefore provides an observable reference for the local queue environment without requiring SHADOW-PPOV to predict that environment.

Order-flow competition and routing. Foucault–Menkveld [20] analyze competition for order flow across venues and the routing systems that arbitrate it, and Laruelle–Lehalle–Pagès [21] give stochastic-approximation algorithms that learn the optimal split of an order across liquidity pools. Where this literature *solves* for the venue split, SHADOW-PPOV’s split *emerges* from placing on the originating venue of each followed order (Section 6.7).

Practitioner references. The practical construction and calibration of VWAP/TWAP/POV and their trade-offs are documented in practitioner texts such as Kissell [5], which we take as the reference description of the methods in Section 2.

Recent execution research. Contemporary execution research is largely forecast- and optimization-based, which sharpens the contrast with SHADOW-PPOV. Reinforcement-learning formulations place market and limit orders to minimize implementation shortfall, both in general limit-order settings [24] and within queue-reactive order-book models [23], and signal-adaptive schemes optimize quote placement from predictive features. Optimal execution under *passive* market impact [22] models fill and impact as functions of spread, queue position, displayed depth, imbalance, and recent order flow — the same state SHADOW-PPOV does not compute. Work on the market maker’s trade-off between fill probability and post-fill return [25] quantifies the top-of-book adverse selection that any resting order, including a shadow, inherits (Section 10). SHADOW-PPOV differs from all of these in computing no forecast and no optimization: it places in reaction to observed resting orders.

4 Prior patents

Four patented mechanisms sit closest to SHADOW-PPOV. Each shares a surface resemblance — passivity, pegging, or following — but differs in a way that is material to SHADOW-PPOV’s design.

- **Pegged / tracking orders** (e.g. US 10,614,520, “Tracking Liquidity Order” [26]). A non-displayed order pegs to the same side of the National Best Bid and Offer (NBBO), continuously *re-pricing against evaluated market state*. SHADOW-PPOV evaluates no such state to manage its shadow: price is inherited once from the shadowed add, and the shadow is *cancelled* when that add ends.
- **Copy / mirror trading** (e.g. US 2013/0268423, “Copy Trading System and Method” [27]). These replicate *executed trades* of *identified* accounts a user has chosen to follow. SHADOW-PPOV follows *anonymous passive adds* at the message level — the posting of liquidity itself — and couples both entry *and cancellation* to the followed order’s identifier.

- **Exchange-side passive/hidden liquidity** (e.g. US 2006/0253379, “Passive Liquidity Order”). This is a matching-engine mechanism internal to a venue; SHADOW-PPOV instead is a client-side strategy that observes a public feed and places and cancels its own orders.
- **Hidden-liquidity detection** (e.g. US 8,140,416). Here order identifiers are tracked for *analysis* (inferring hidden size), and trading decisions still evaluate market state; SHADOW-PPOV uses identifier presence directly as the trigger for a cancellation.

5 Positioning relative to prior work

Having reviewed the existing execution methods (Section 2), the academic literature (Section 3), and the closest patented mechanisms (Section 4), we state where SHADOW-PPOV sits relative to all three. The contribution is not the general idea of following displayed liquidity or using participation rates, which are established in both the public literature and prior patents. It is the specific combination of order-level shadowing, identifier-keyed lifecycle mirroring, and same-venue placement as a model-free passive-placement benchmark — a combination of design choices that all follow from one commitment, to build no model of the book and follow observed orders instead:

1. **Model-free placement.** The passive order’s price is inherited from the observed add it follows, so placement needs no order-book model and no fill-probability prediction; this is what lets SHADOW-PPOV serve as a benchmark for predictive placement models (Section 10).
2. **Identifier-keyed cancellation.** A shadow-order is withdrawn when a delete or trade message for the specific order it follows arrives, rather than when evaluated market state (spread, queue decay, fill probability) crosses a threshold.
3. **Flow-inherited venue selection.** Each shadow is sent to the originating venue of the add it follows, so venue choice is a pass-through rather than an optimization; the cross-venue mix of executions thereby tracks, in expectation, the cross-venue mix of observed liquidity provision without explicit routing logic — the inverse of the SOR approach [20, 21], where the split is solved for.
4. **Constant per-message decision cost.** With only a single identifier association retained per followed order, each message is handled by one identifier lookup, so the per-message decision cost is $O(1)$ expected time, independent of the number of tracked shadows.

Table 2 summarizes the positioning.

6 The Shadow-PPOV Method

6.1 Message model

SHADOW-PPOV consumes a real-time stream of order-book messages from one or more venues (e.g. an ITCH-style binary feed). Each message carries a *message type* — ADD (a new resting limit order), DELETE (cancellation), TRADE (execution), MODIFY (amendment) — an *exchange-assigned order identifier*, and an *originating venue identifier*, together with side/price/quantity for ADD/MODIFY. Individual order lifecycles are therefore followable at the message level by identifier.

Table 2: How SHADOW-PPOV differs from the nearest prior work.

Prior work	Mechanism	How SHADOW-PPOV differs
Almgren–Chriss [1]	Optimal <i>schedule</i> from a volume/impact model	No schedule; participation emerges from following individual adds
POV [5]	Reactive participation to <i>aggregate</i> realized volume; leans ahead to stay passive	Follows <i>individual</i> resting orders; never posts ahead
Moallemi–Yuan queue value [19]	Quantifies queue-position adverse selection	Does not lead the queue; inherits a resting order’s neighborhood
SOR [20, 21]	Computes or learns the venue split	Split <i>emerges</i> from same-venue placement; no routing computed
Tracking/pegged orders [26]	Re-prices against evaluated NBBO state	No state evaluated; cancels on followed-order termination
Copy/mirror trading [27]	Copies executed trades of identified accounts	Follows anonymous passive <i>adds</i> ; keys cancellation to their identifier

6.2 Core mechanic: participant shadowing by identifier

On each ADD that satisfies a selection criterion (below), SHADOW-PPOV transmits a *shadow* limit order at the add’s price to the add’s originating venue, and receives a venue-assigned shadow-order identifier. It then records a single association keyed by the observed order’s (venue, `exchange_order_id`) — exchange order identifiers are unique only within a venue — and mapping it to its shadow-order identifier,

$$(\text{venue}, \text{exchange_order_id}) \mapsto \text{shadow_order_id}.$$

The shadow is thereafter *bound* to the observed order by that key: it inherits that order’s price and venue, and — crucially — its *lifecycle*. The single invariant that makes this “shadowing” rather than “quoting” is this attachment; a shadow is a projection of one identified resting order.

6.3 Order-placement rate

SHADOW-PPOV shadows a bounded fraction of qualifying adds. The reference implementation offers two independent selection modes:

- *Probabilistic*: shadow a qualifying add with a fixed probability ρ (draw a pseudorandom value and compare to a threshold; the reference implementation uses $\rho = 2/128 \approx 1.5\%$, a power-of-two modulus so the test compiles to a mask rather than a division).
- *Deterministic*: shadow every N -th qualifying add (reference-implementation default $N = 5$, i.e. a 20% selection rate).

The two modes give different selection rates; the experiments reported in Section 8 use the probabilistic mode at $\rho \approx 1.5\%$. The quantity SHADOW-PPOV sets is thus the *order-placement rate* ρ — the fraction of qualifying adds it shadows — and *not* a participation rate. This placement rate is fixed by construction and applied to observed order flow, not to traded volume; SHADOW-PPOV places more shadows when adds are frequent and fewer when they are sparse. The realized *participation* — SHADOW-PPOV’s share of traded volume — cannot be controlled directly; it emerges from following at the fixed placement rate. Because not every placed shadow fills, the realized participation is

not equal to the order-placement rate and will generally differ from it. Participation is therefore a random quantity whose dependence on parent-order size is analyzed in Section 9.

6.4 Shadow quantity

The shadow’s *desired* quantity is computed independently of the triggering add’s quantity — in the reference implementation as a randomized function of average trade sizes over a preceding window. Before transmission the desired quantity is capped by the followed order’s size, the per-order and per-level limits (e.g. 1-lot child orders with a 25-lot aggregate ceiling), and the remaining parent quantity. Fills follow from co-resting at the same price: a sufficiently large sweep can consume the followed order and continue into the shadow, while a partial sweep may stop before reaching the shadow.

6.5 Retained information

For each followed order, the core association retains *only* its (venue, exchange identifier) key paired with the shadow identifier; it does not retain the followed order’s price, quantity, queue position, order-book state, or timestamp. The algorithm additionally maintains global configuration — target side, selection rate, and grace-window parameters — together with the minimal state needed to manage its own orders. Two consequences follow. First, no price/queue/state-based cancellation decision is *possible*, because the inputs to such a decision are not retained; cancellation is forced to be identifier-driven by construction. Second, storing, looking up, and removing an association by identifier is expected $O(1)$, so the cost of the cancellation decision is independent of the number of tracked orders.

6.6 Identifier-driven cancel mirroring

On a lifecycle-terminating message for a followed order, SHADOW-PPOV looks up the message’s exchange identifier. If no association is present, nothing happens. If it is present, the response depends on *how* the followed order ended:

- **Pulled (delete).** The liquidity simply left; no aggression is in flight to fill against. SHADOW-PPOV cancels the shadow *immediately* and removes the association — continuing to rest there would be pure adverse-selection exposure.
- **Hit (trade).** A trade is a marketable sweep still in flight, and because the shadow rests in the same queue behind the followed order, that sweep may be about to reach it. Cancelling instantly would pull the shadow out of the queue *before* the aggression arrives — and in the worst case the shadow would then *never* fill, because it always cancels in time. SHADOW-PPOV therefore arms a *delayed cancel*: it holds the shadow live for a short grace window of B end-of-burst events, during which the shadow may itself be filled by the continuing sweep. If the window elapses without a fill, the cancel is sent and the association removed. Arming is idempotent: a further trade for the same order while the window is armed does not restart it.

Aside from this trade-versus-pull distinction, the cancellation input is still just identifier presence: no book depth, queue position, price competitiveness, or fill probability is evaluated. The hard safety guards of the reference implementation (Section 7) — target reached, over-hedge, or price drifted past the cancel band — may fire an *immediate* cancel that pre-empts an armed grace window. A shadow therefore never outlives the order it follows by more than the grace window, preserving the

Algorithm 1: SHADOW-PPOV identifier-driven core (per message)

state: mapping $M : k \mapsto \text{shadow_id}$ with venue-scoped key $k = (\text{venue}, \text{ex_id})$ (exchange ids are unique only within a venue); target side σ ; order-placement rate ρ ; grace window B bursts; own-id set O of $(\text{venue}, \text{shadow_id})$ pairs

Input: message m with type $m.\tau$, exchange id $m.\text{id}$, venue $m.v$, side $m.s$, price $m.p$; write $k = (m.v, m.\text{id})$

```
1 switch  $m.\tau$  do
2   case ADD do
3     if  $k \in O$  or  $m.s \neq \sigma$  then return // self / wrong side
4     if selection criterion not met (prob.  $\rho$  / every- $N$ ) then return
5      $\text{shadow\_id} \leftarrow$  transmit shadow at price  $m.p$  to venue  $m.v$ ;
6      $O \leftarrow O \cup \{(m.v, \text{shadow\_id})\}$ ;  $M[k] \leftarrow \text{shadow\_id}$ 
7   case DELETE (followed order pulled) do
8     if  $k \in M$  then cancel  $M[k]$  now; remove  $M[k]$ 
9   case TRADE (followed order hit) do
10    if  $k \in M$  and not already armed then
11      arm delayed cancel of  $M[k]$  in  $B$  bursts // grace: the sweep may still
      fill the shadow
12  case MODIFY do
13    if  $k \in M$  then
14      cancel  $M[k]$  now; remove  $M[k]$  // teardown

// grace-window expiry or safety guard  $\Rightarrow$  cancel armed shadow; own fill  $\Rightarrow$ 
// remove its association
// on parent complete: cancel all shadows in  $M$ ; clear  $M$  (wind-down)
```

discipline that SHADOW-PPOV rests only where a currently-live participant order rests, or rested an instant ago. A *MODIFY* for a tracked order is treated as termination of the followed lifecycle: the shadow is cancelled and the association removed. Re-shadowing is not automatic — the amended order is shadowed anew only if the feed subsequently emits a fresh qualifying *ADD* for it. This is an algorithmic convention rather than an assertion about exchange modify semantics.

6.7 Flow-inherited venue selection

Because each shadow is transmitted to the originating venue of the add it follows, a fixed-rate selector preserves, in expectation, the cross-venue distribution of qualifying observed adds. If participants rest 50%/30%/20% of adds on venues A/B/C, a fixed selection rate reproduces that 50:30:20 mix in expectation — *without* venue liquidity metrics, fee/rebate optimization, latency-based routing, or any explicit routing logic. This is flow-inherited venue selection, not an optimization: the venue mix is proportional to the observed distribution of qualifying adds, which need not coincide with the split that minimizes fees, latency, or adverse selection, nor with the distribution of traded volume.

7 Reference Implementation: the kaspar-hft System

KASPAR-HFT is an open-source, low-latency trading system written in C++. It is built using the actor-model architecture [3, 4]: independent components communicate only by passing messages, so concurrent order handling stays deterministic and keeps shared-memory locking off the critical path. The same code runs unchanged across backtesting, market replay, live simulation, and live trading; the evaluation of Section 8 therefore exercises the live code path directly.

SHADOW-PPOV is implemented as a set of execution components within KASPAR-HFT. A strategy issues a position order, and the execution components translate it into the individual limit orders sent to the exchange, with the acquiring and liquidating sides of each instrument handled independently. This implementation is a *superset* of Algorithm 1: it keeps the identifier-driven attach/cancel core and adds state-based *safety guards* appropriate to running real capital against a live venue.

The added guards are:

- **Two-band place/cancel hysteresis.** A shadow is only *placed* when its price is within a tight placement band δ_{place} ticks of the inside (e.g. for a buy, $p \geq b_0 - \delta_{\text{place}}$ against best bid b_0), but is only *cancelled for distance* once it drifts beyond a looser cancel band δ_{cancel} , with $\delta_{\text{place}} < \delta_{\text{cancel}}$ (reference implementation: 4 and 6 ticks). The gap introduces hysteresis: it prevents a one-tick oscillation of the inside from triggering a place/cancel/replace loop that would burn message budget and queue priority. The placement band δ_{place} also bounds how far from the inside a shadow can attach; tightening it restricts shadowing to adds near the touch, which fill more reliably and reduce the small-order participation dispersion of Section 9.
- **Completion and over-hedge guards.** A working shadow is also pulled when the position reaches its target or goes flat, when the size resting at its price would overshoot the quantity that remains to be executed, or when per-level order caps are exceeded. The placed size is the minimum of the per-level cap, the per-order cap, the remaining quantity, and the followed order’s size, clamped at zero. These guards make the shadow never work more than remains and never post more than the participant it follows.

Grace window. As described in Section 6.6, the shadow’s cancel is delayed only when the followed order is hit by a trade, to give the incoming sweep a chance to fill the shadow; when the followed order is instead cancelled, the shadow is cancelled at once. The safety guards override this delay: whenever a completion, over-hedge, or distance limit is reached, the shadow is cancelled immediately.

In short, the core method is the minimal identifier-driven algorithm (Algorithm 1); the reference implementation runs that core plus the guards above. Guarded cancellations are a strict addition to the identifier-keyed cancel. These guards read market state — the inside price and recent trade sizes — so the production wrapper is not itself model-free; the model-free property is a claim about the core placement and cancellation logic of Algorithm 1, not about the safety layer around it.

7.1 Deterministic replay simulator

The KASPAR-HFT simulator replays recorded exchange packet captures through this same code. It reconstructs the book from the captured feed and places each shadow into the price-time-priority (FIFO) queue at its price, behind the depth already resting there. Under this reconstructed queue, the simulator assigns a shadow a fill when subsequent recorded executions imply that aggression has reached a resting order *behind* it; because the shadow has earlier time priority than that order,

the simulator treats the shadow as filled as well. This is a simulator inference from the recorded feed, not a direct observation of the synthetic order.

The execution algorithm and the order-book simulator used in this evaluation are available as the open-source KASPAR-HFT project, which supports testing and live deployment against CME or other venues.

8 Experimental Evaluation

8.1 Setup

We evaluate SHADOW-PPOV on execution of a two-leg index-futures spread (NQ against ES) on CME. Each session *fires* (activates) the parent in ten lock-step chunks, twice per date, at two times: 09:29 ET (open) and 15:57 ET (close). Running both directions on the same date yields a *direction-agnostic* slippage estimate in which the mid-price reference cancels algebraically (below).

Corpus. The corpus is 137 date-pairs (one long/short pair per session) over US trading dates from 2025-06-01 to 2026-02-01, two runs per date; we use “session” and “date-pair” interchangeably. Front-month contracts follow the CME quarterly-roll calendar, spanning the ESM5/NQM5 through ESH6/NQH6 fronts.

Slippage definition. For a leg, let mid be its best bid and offer (BBO) midpoint cached at the instant the chunk alarm fires, and vwap the size-weighted average execution price for that leg on that direction/date. Define the size-weighted, direction-agnostic per-contract slippage

$$\text{cost}_{\text{BUY}} = \text{vwap}_{\text{BUY}} - \text{mid}, \quad (1)$$

$$\text{cost}_{\text{SEL}} = \text{mid} - \text{vwap}_{\text{SEL}}, \quad (2)$$

$$\text{Slippage} = \frac{1}{2}(\text{cost}_{\text{BUY}} + \text{cost}_{\text{SEL}}) = \frac{1}{2}(\text{vwap}_{\text{BUY}} - \text{vwap}_{\text{SEL}}). \quad (3)$$

In the second equality the fire-time mid cancels, so the reported figure is exactly *half the gap between what the buyer of a leg paid and what the seller of that leg received*, independent of which mid was captured. Positive = a loss (buyer paid above mid / seller received below); a negative value flags that the two direction-runs saw different market states.

Units. One bid-ask width (b/o) is one tick = 0.25 points. For NQ, $0.25\text{pt} \times \$20/\text{pt} = \5.00 per b/o per contract; for ES, $0.25\text{pt} \times \$50/\text{pt} = \12.50 per b/o per contract.

8.2 Results

Table 3 reports pooled figures by fire time and product.

Close cohort. At the close the two direction-runs fill against essentially the same market state, with less intraday drift than at the open, and the (fill – mid) framing of Eq. (3) isolates a per-contract slippage. NQ closes at 0.56 b/o = \$2.80 per contract (19,540 contracts) and ES at 0.45 b/o = \$5.59 per contract (31,354 contracts). Both are positive: SHADOW-PPOV executes slightly worse than mid, consistent with adverse selection and execution against a moving market.

Table 3: Direction-agnostic per-contract slippage over 137 date-pairs. Positive = a loss (worse than mid). The close rows are used in the reported results; the open rows are affected by intraday drift (see text).

Phase	Product	Contracts	Slippage (b/o)	Slippage (\$)
open	NQ	20,921	+1.76	+\$8.79
open	ES	36,550	+0.22	+\$2.76
close	NQ	19,540	+0.56	+\$2.80
close	ES	31,354	+0.45	+\$5.59

Open cohort. Near the open the price moves sharply: over 100 NQ points within the 60-second window, and 171 points (684 ticks) on one date. Across that move the two direction-runs sample different price levels, so the open per-contract slippage reflects that price drift as well as execution — in one case it is even negative, driven by the inter-run drift rather than by execution. The conclusions in this paper are based on the close cohort.

8.3 Threats to validity

The evaluation is a *replay* study, and its one material limitation is that the simulator does not model the market’s reaction to SHADOW-PPOV’s own orders: each shadow is filled against the recorded feed as though its presence did not perturb the flow it follows. For the small sizes and order-placement rate used ($\rho \approx 1.5\%$) the study does not independently validate the zero-impact assumption; it would break once a shadow becomes a material fraction of resting liquidity, so the reported figures should not be extrapolated to sizes that large. The evaluation is also narrow in scope — a single two-leg index-futures spread, passive placement only, at a small order-placement rate ($\rho \approx 1.5\%$) — so the numbers should not be read as generalizing to other products, venues, or larger sizes; multi-venue equity evaluation is left to future work.

9 Participation Rate and Parent-Order Size

SHADOW-PPOV controls its *order-placement rate* (Section 6.3); it does not control the participation rate directly, nor the fill of any one shadow. Because each shadow joins the queue behind the add it attaches to, while the amount of depth ahead of it is unknown at placement, whether and when it fills is a random event. A small parent has few fill opportunities, so its realized participation is highly variable; as the number of completed child fills grows, averaging reduces that variability. This is a general property of passive percentage-of-volume and volume-tracking execution rather than a feature specific to SHADOW-PPOV: the delivered participation is a realized quantity, a recognized source of tracking error and execution risk [5, 6, 7, 8]. SHADOW-PPOV inherits this behavior, with the queue-position uncertainty of the followed order as the concrete source of per-fill randomness [19, 15]; the dependence of that dispersion on parent-order size is what the model below quantifies.

A stylized model. To isolate the dependence on size, consider executing a parent of Q one-lot children. Each child attaches to a random add whose queue position is unknown. For illustration we assume the market volume that must trade before a child fills is $\text{Exponential}(\rho)$ with mean $1/\rho$ for a target participation rate ρ (within this stylized model ρ denotes the participation rate, distinct from the order-placement rate of Section 6.3). The market volume elapsed when the parent completes its

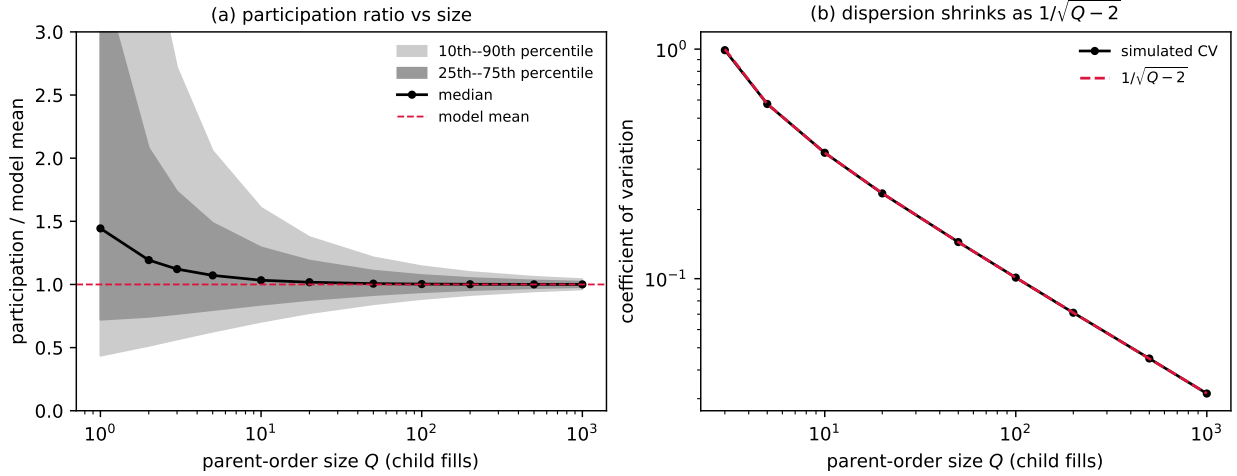


Figure 1: Realized participation relative to the model’s mean versus parent-order size Q (number of one-lot child fills), from the stylized model $\hat{p}/\rho = Q/\text{Gamma}(Q, 1)$. (a) Median and percentile bands: the ratio is dispersed for small Q and concentrates on the model’s mean (dashed line) as Q grows. (b) The coefficient of variation falls as $1/\sqrt{Q-2}$.

Q fills is

$$M = \sum_{i=1}^Q G_i, \quad G_i \sim \text{Exponential}(\rho) \implies M \sim \text{Gamma}(Q, \rho),$$

and the realized participation is $\hat{p} = Q/M$. The ratio $\hat{p}/\rho = Q/\text{Gamma}(Q, 1)$ is independent of ρ , so the dependence on ρ cancels and the model predicts the same normalized size dependence for any ρ . The qualitative conclusion — concentration on the model’s mean as Q grows — follows from the central limit theorem and holds for a broad class of finite-variance gap distributions. The independent-exponential gap assumption is a strong simplification: real fill gaps are correlated across concurrent shadows, volatility regimes, and clustered aggressive events, and modeling the children as filling one at a time produces greater participation variance than allowing multiple concurrent fills to average independently. The model does not estimate SHADOW-PPOV’s actual participation — the exponential fill-gap distribution and its rate are not fitted to the CME corpus — so the result is a scaling law under a stylized model, not an empirical claim that SHADOW-PPOV achieves a particular participation rate.

Results. Figure 1 and Table 4 report a Monte Carlo of $\hat{p}/\rho$ over 4×10^5 trials per size. At $Q = 1$ the ratio is $1/\text{Exponential}(1)$: heavy-tailed with a divergent mean and a 10th–90th percentile band spanning $[0.44, 9.5]$ times the mean — the participation is effectively random. As Q grows the ratio concentrates on the model’s mean and its coefficient of variation falls as $1/\sqrt{Q-2}$ (asymptotically $1/\sqrt{Q}$): the 10th–90th band narrows to $[0.70, 1.61]$ at $Q = 10$, $[0.89, 1.14]$ at $Q = 100$, and $[0.96, 1.04]$ at $Q = 1000$.

Implication. The size at which realized participation becomes concentrated bounds where SHADOW-PPOV, or any passive follower, delivers a stable participation. Under the stylized model a parent comprising hundreds or thousands of child fills has realized participation within a few percent of the model’s mean; a very small order does not, and its participation is effectively random. For a small parent order the dispersion can be reduced by modifying SHADOW-PPOV to shadow

Table 4: Participation relative to the model’s mean versus parent-order size, 4×10^5 Monte Carlo trials per size. CV is finite only for $Q > 2$.

Q (child fills)	median	10th pct	90th pct	CV
1	1.44	0.44	9.48	—
2	1.19	0.51	3.77	—
5	1.07	0.63	2.06	0.58
10	1.03	0.70	1.61	0.35
20	1.02	0.77	1.38	0.24
50	1.01	0.84	1.22	0.14
100	1.00	0.89	1.14	0.10
200	1.00	0.92	1.10	0.07
500	1.00	0.95	1.06	0.05
1000	1.00	0.96	1.04	0.03

only adds reasonably close to the inside of the book (tightening the placement band of Section 7), which have a higher and more predictable fill probability; without such a restriction a small order’s participation is dominated by the queue-position randomness of the few orders it follows, as the small- Q case shows. This also refines the benchmark claim of Section 10: SHADOW-PPOV is a benchmark for passive placement at sizes large enough that the participation band is narrow, and the width of the participation band is a second axis — alongside slippage — on which a predictive placement model that controls queue position can be scored against it.

10 A benchmark for passive placement

SHADOW-PPOV places passive orders without an order-book model and without predicting where they will fill, and it executes at the slippage measured in Section 8. We position it as a benchmark for passive-order placement on a CLOB rather than as an optimal method. Given the same passive mandate, a predictive placement model can be scored against SHADOW-PPOV to test whether its additional modeling produces a measurable improvement; the present study reports no such competing baselines, so the benchmark role is proposed rather than demonstrated.

Adaptive scheduling methods are already evaluated against fixed references: a reinforcement-learning or optimal-execution scheduler is judged against TWAP, VWAP, and the Almgren–Chriss trajectory [1, 23], so any claimed gain is stated relative to a model-free baseline. A comparable model-free reference is useful when evaluating the additional complexity of a predictive placement model. The naive alternatives — posting at the touch, or post-and-wait then cross [14, 17] — are ad hoc; SHADOW-PPOV offers a well-specified alternative. A practitioner proposing a predictive placement model can show whether it beats shadow-following, as an execution scheduler is judged against TWAP.

SHADOW-PPOV is an execution method: its scope is completing a given position at a small, predictable cost, with a small number of decision and state operations per message. When direction is known, an informed aggressive strategy outperforms it, and SHADOW-PPOV leaves such directional decisions to the strategy layer above it.

11 Future Work

The measurements of Section 8 are preliminary. They establish the mechanism and give a first replay estimate of its slippage, not a completed benchmark study, and several directions remain before SHADOW-PPOV can be treated as a validated reference.

Comparison against established methods. The central open task is to score SHADOW-PPOV against competing passive-placement rules under the same direction-neutral slippage metric: posting at the touch; a random same-level “null” follower, which isolates whether following a *specific* identified order carries information beyond posting behind recently added liquidity; the passive legs of TWAP, VWAP, and POV; and at least one *predictive* placement model. Only such a comparison can establish whether SHADOW-PPOV is a useful benchmark and by how much a predictive model improves on it.

Understanding the algorithm’s behavior. Beyond a single cost figure, the method’s behavior needs characterization: realized fill and completion rates, the fraction of followed orders whose lifecycle ends in a fill rather than a cancellation, queue-position dynamics, and sensitivity to the order-placement rate ρ and the grace window B .

Statistical strength. The reported figures should be accompanied by confidence intervals, trade and parent-order counts, and per-session dispersion, and the safety guards (the two-band hysteresis and the completion and over-hedge limits) should be ablated to quantify their effect on the reported costs.

Market impact and breadth. The replay uses a zero-impact simulator; a controlled experiment that injects synthetic size and measures the perturbation of subsequent flow would test that assumption. The corpus is one two-leg index-futures spread; extending to other products, to fragmented multi-venue equity markets — where the flow-inherited venue selection of Section 6.7 would actually be exercised — and to larger sizes is required before the results can be generalized.

12 Conclusion

We presented SHADOW-PPOV, a passive execution method that trades the schedules and aggregate-volume targets of VWAP/TWAP/POV for direct observation of the passive orders participants actually post. By shadowing individual resting orders by exchange identifier — mirroring price and venue, and tying cancellation to the shadowed order’s lifecycle through a lookup on the followed order’s identifier — SHADOW-PPOV needs no volume forecast of its own, uses followed orders as observable references for local execution flow, obtains venue selection as an emergent property rather than a routing computation, and makes cancellation an identifier-driven expected- $O(1)$ decision. Over 137 replayed CME sessions, the close-cohort slippage is \$2.80/contract (NQ) and \$5.59/contract (ES) — slightly worse than mid, consistent with a passive fill against a moving market. Because it places passive orders with no order-book model and no fill prediction yet its replayed close-cohort executions remain within roughly half a bid-ask width of mid, we propose SHADOW-PPOV as a model-free benchmark for passive-order placement — a reference a predictive placement model should be expected to beat. The method and its identifier-driven cancel-mirroring mechanism are the subject of a pending patent application.

Reproducibility and disclosure

Results are produced by a deterministic replay of recorded CME PCAP market data through open-source C++ actor code (Section 7.1); a fixed session replays to identical fills. The SHADOW-PPOV implementation and the order-book simulator are available as the KASPAR-HFT project.²

Acknowledgments. The slippage-measurement corpus and the deterministic replay harness were built at M2 Technologies.

References

- [1] R. Almgren and N. Chriss. *Optimal Execution of Portfolio Transactions*. Journal of Risk, 3(2):5–40, 2001.
- [2] M. Avellaneda and S. Stoikov. *High-Frequency Trading in a Limit Order Book*. Quantitative Finance, 8(3):217–224, 2008.
- [3] C. Hewitt, P. Bishop, and R. Steiger. *A Universal Modular ACTOR Formalism for Artificial Intelligence*. Proc. 3rd International Joint Conference on Artificial Intelligence (IJCAI), 1973.
- [4] G. Agha. *Actors: A Model of Concurrent Computation in Distributed Systems*. MIT Press, 1986.
- [5] R. Kissell. *The Science of Algorithmic Trading and Portfolio Management*. Academic Press, 2013.
- [6] Á. Cartea, S. Jaimungal, and J. Penalva. *Algorithmic and High-Frequency Trading*. Cambridge University Press, 2015.
- [7] J. Białkowski, S. Darolles, and G. Le Fol. *Improving VWAP Strategies: A Dynamic Volume Approach*. Journal of Banking & Finance, 32(9):1709–1722, 2008.
- [8] C. Frei and N. Westray. *Optimal Execution of a VWAP Order: A Stochastic Control Approach*. Mathematical Finance, 25(3):612–639, 2015.
- [9] B. Johnson. *Algorithmic Trading & DMA: An Introduction to Direct Access Trading Strategies*. 4Myeloma Press, London, 2010.
- [10] *Bayesian Trading Cost Analysis and Ranking of Broker Algorithms*. arXiv:1904.01566, 2019.
- [11] A. F. Perold. *The Implementation Shortfall: Paper versus Reality*. Journal of Portfolio Management, 14(3):4–9, 1988.
- [12] Á. Cartea and S. Jaimungal. *Optimal Execution with Limit and Market Orders*. Quantitative Finance, 15(8):1279–1291, 2015.
- [13] O. Guéant, C.-A. Lehalle, and J. Fernández-Tapia. *Optimal Portfolio Liquidation with Limit Orders*. SIAM Journal on Financial Mathematics, 3(1):740–764, 2012. [arXiv:1106.3279](https://arxiv.org/abs/1106.3279).
- [14] R. Cont and A. Kukanov. *Optimal Order Placement in Limit Order Markets*. Quantitative Finance, 17(1):21–39, 2017. [arXiv:1210.1625](https://arxiv.org/abs/1210.1625).

²<https://github.com/vincent212/kaspar-hft>

- [15] T. DeLise. *The Negative Drift of a Limit Order Fill*. arXiv:2407.16527, 2024.
- [16] *Design and Implementation of Schedule-Based Trading Strategies Based on Uncertainty Bands*. arXiv:1409.1441, 2014.
- [17] *On the Design of Sell-Side Limit and Market Order Tactics*. arXiv:1409.1442, 2014.
- [18] R. Cesari et al. *Effective Trade Execution*. arXiv:1206.5324, 2012.
- [19] C. C. Moallemi and K. Yuan. *A Model for Queue Position Valuation in a Limit Order Book*. Working paper, Columbia University, 2016. SSRN 2996221.
- [20] T. Foucault and A. J. Menkveld. *Competition for Order Flow and Smart Order Routing Systems*. Journal of Finance, 63(1):119–158, 2008. doi:[10.1111/j.1540-6261.2008.01312.x](https://doi.org/10.1111/j.1540-6261.2008.01312.x).
- [21] S. Laruelle, C.-A. Lehalle, and G. Pagès. *Optimal Split of Orders Across Liquidity Pools: A Stochastic Algorithm Approach*. SIAM Journal on Financial Mathematics, 2(1):1042–1076, 2011. arXiv:0910.1166.
- [22] A. Barzykin, R. Boyce, E. Neuman, and S. Tuschmann. *Optimal Execution with Passive Market Impact*. arXiv:2607.28323, 2026.
- [23] T. Espana, Y. Hafsi, F. Lillo, and E. Vittori. *Reinforcement Learning in Queue-Reactive Models: Application to Optimal Execution*. arXiv:2511.15262, 2025.
- [24] P. Cheridito and M. Weiss. *Reinforcement Learning for Trade Execution with Market and Limit Orders*. arXiv:2507.06345, 2025.
- [25] J. Albers, M. Cucuringu, S. Howison, and A. Y. Shestopaloff. *The Market Maker’s Dilemma: Navigating the Fill Probability versus Post-Fill Returns Trade-Off*. arXiv:2502.18625, 2025.
- [26] *Tracking Liquidity Order*. U.S. Patent No. 10,614,520 B2, 2020. patents.google.com/patent/US10614520B2.
- [27] *Copy Trading System and Method*. U.S. Patent Application Publication No. 2013/0268423 A1 (Pennaluna & Company, Inc.), 2013. patents.google.com/patent/US20130268423A1.
- [28] *Passive Liquidity Order*. U.S. Patent Application Publication No. 2006/0253379 A1, 2006.
- [29] *Algorithmic Trading — Hidden Liquidity*. U.S. Patent No. 8,140,416 B2, 2012.